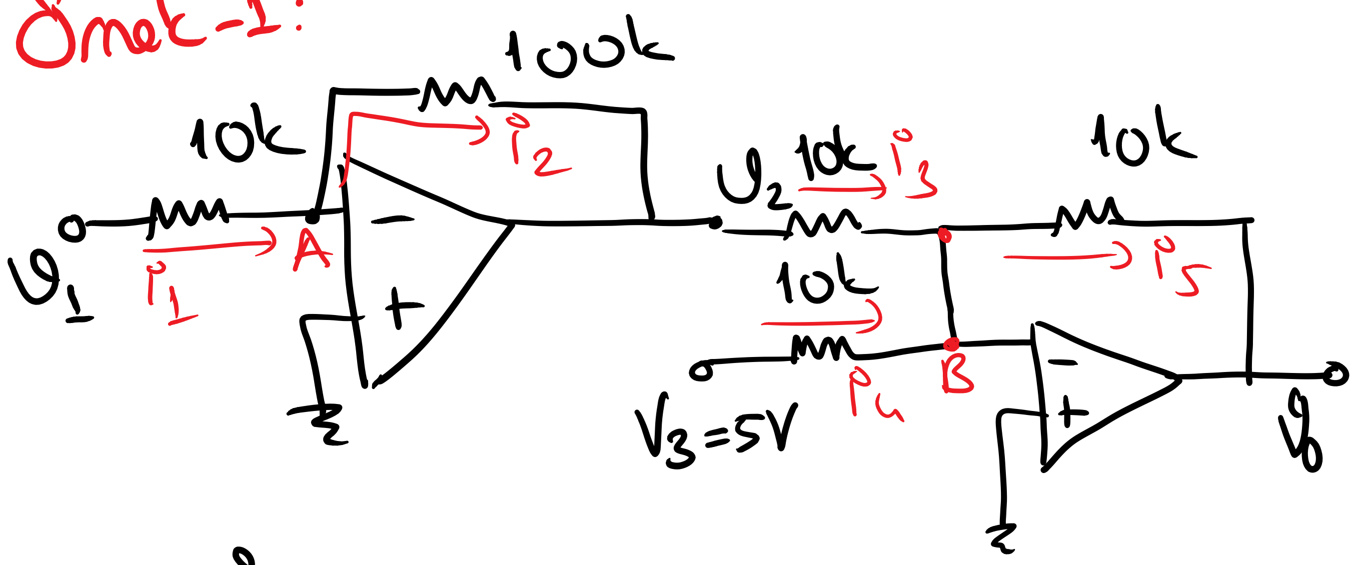
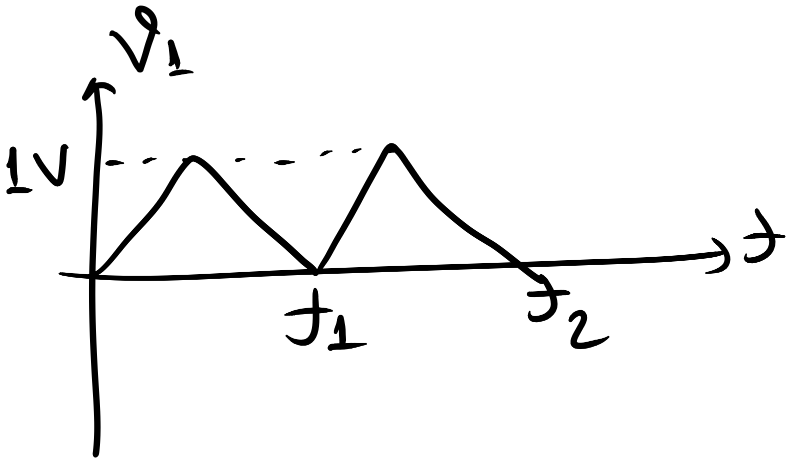
Karışık Örnekler

Örnek-1:





$\pm V_{cc} = \pm 12V$

* 1. op-amp'ta.

$$i_1 = i_2$$

$$\frac{v_1}{10k} = \frac{-v_2}{100k}$$

$$\boxed{v_2 = -10v_1}$$

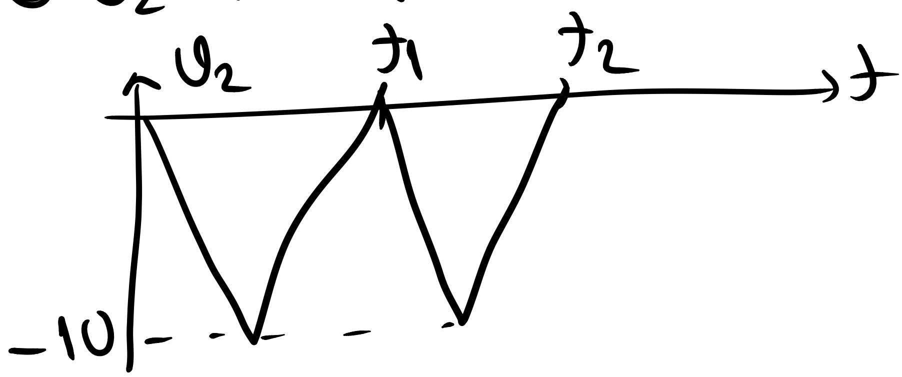
* 2. op-amp'ta

$i_3 + i_4 = i_5$

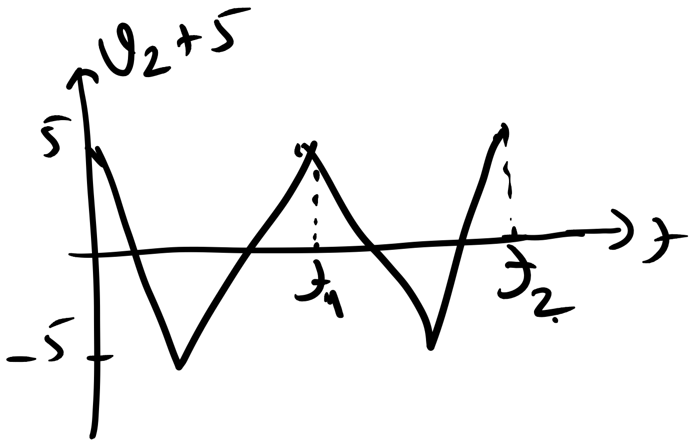
$$\frac{v_2}{10k} + \frac{5}{10k} = \frac{-v_0}{10k} \Rightarrow \boxed{v_0 = -(v_2 + 5)}$$

Çizim:

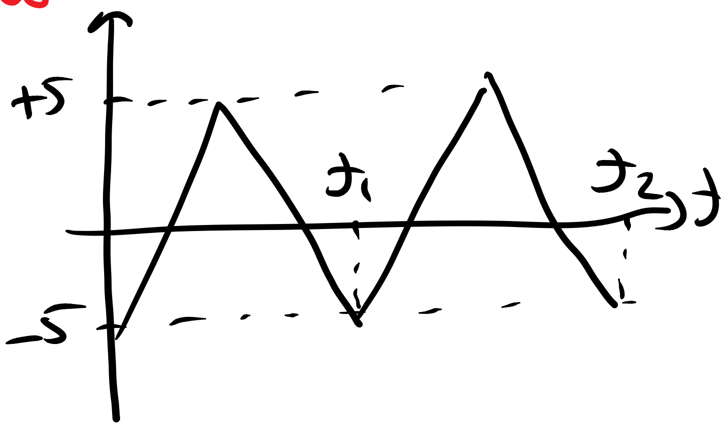
- v_2'nin çizimi



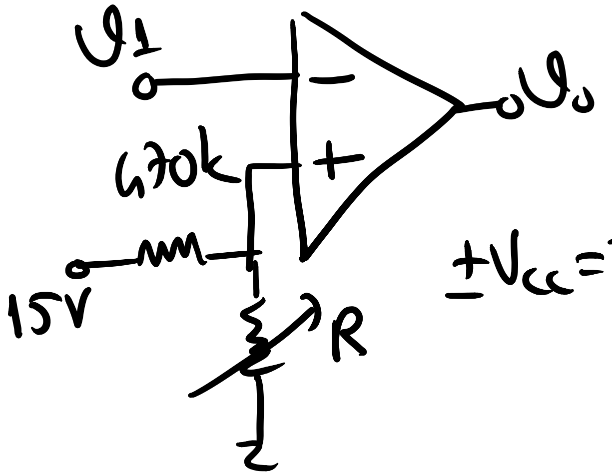
- v_0'ın çizimi:



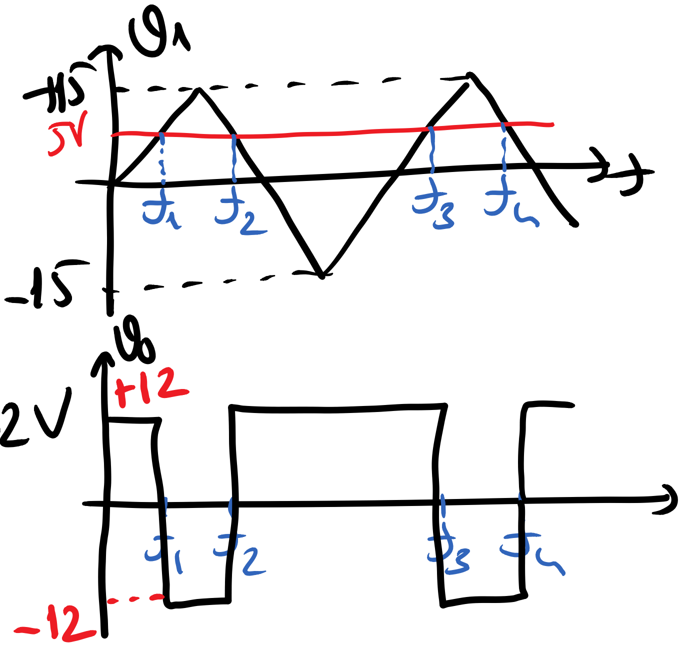
$v_0 = -(v_2+5)$



Örnek-2:



$\pm V_{cc} = \pm 12V$

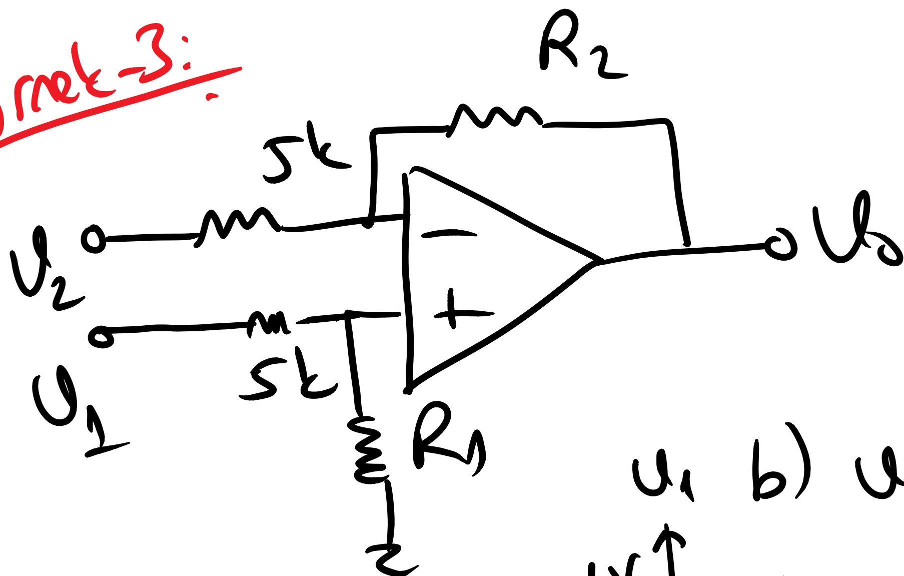


$R = ?$

$$V_+ = 5V \Rightarrow \frac{15}{470k + R} \cdot R = 5$$

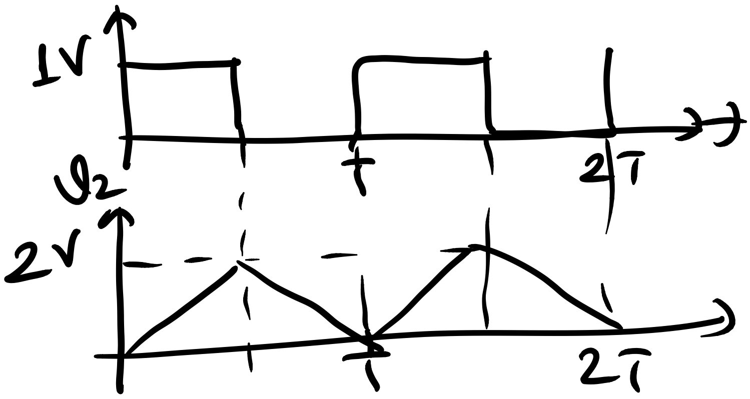
$$\Rightarrow \boxed{R = 235\,k\Omega}$$

Örnek-3:



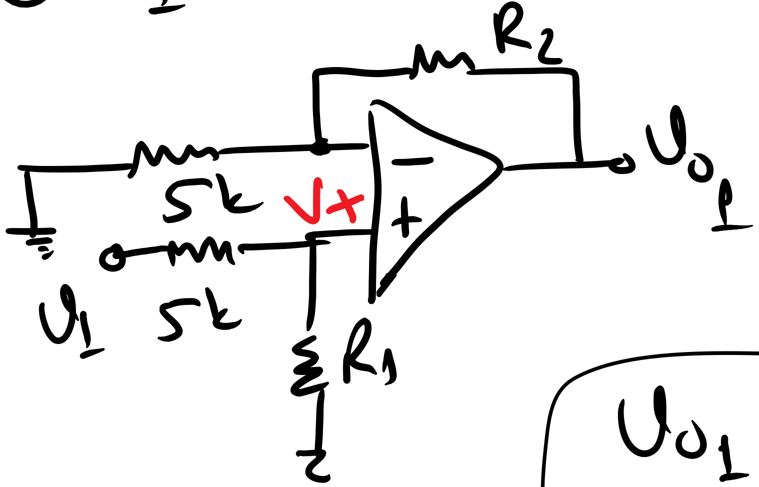
a) $v_0 = 2v_1 - 3v_2$ ise $R_1 = ?$ $R_2 = ?$

b) $v_0 = ?$ çiz



a) süperpozisyon.

* U_1'in etkisi ($U_2 = 0V$)

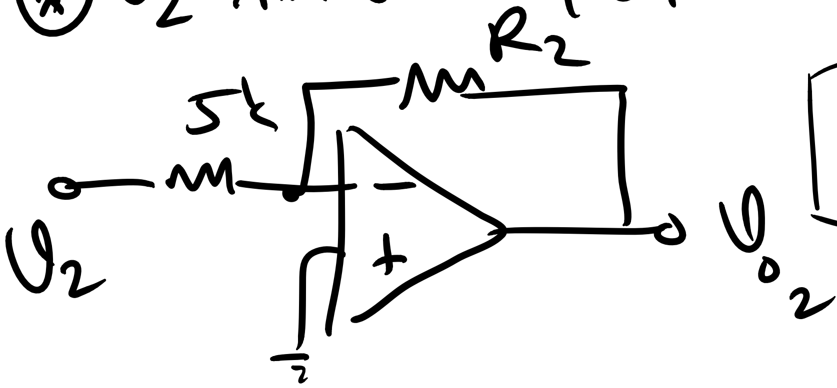


$$U_{o_1} = \left(1 + \frac{R_2}{5k}\right) \cdot V_+$$

$$V_+ = \frac{U_1}{5k + R_1} \cdot R_1$$

$$\boxed{U_{o_1} = \frac{5k + R_2}{5k} \cdot \frac{R_1}{5k + R_1} \cdot U_1}$$

* U_2'nin etkisi ($U_1 = 0\,V \Rightarrow V_+ = 0V$)



$$\boxed{U_{o_2} = -\frac{R_2}{5k} \cdot U_2}$$

$U_o = 2U_1 - 3U_2$ old. göre;

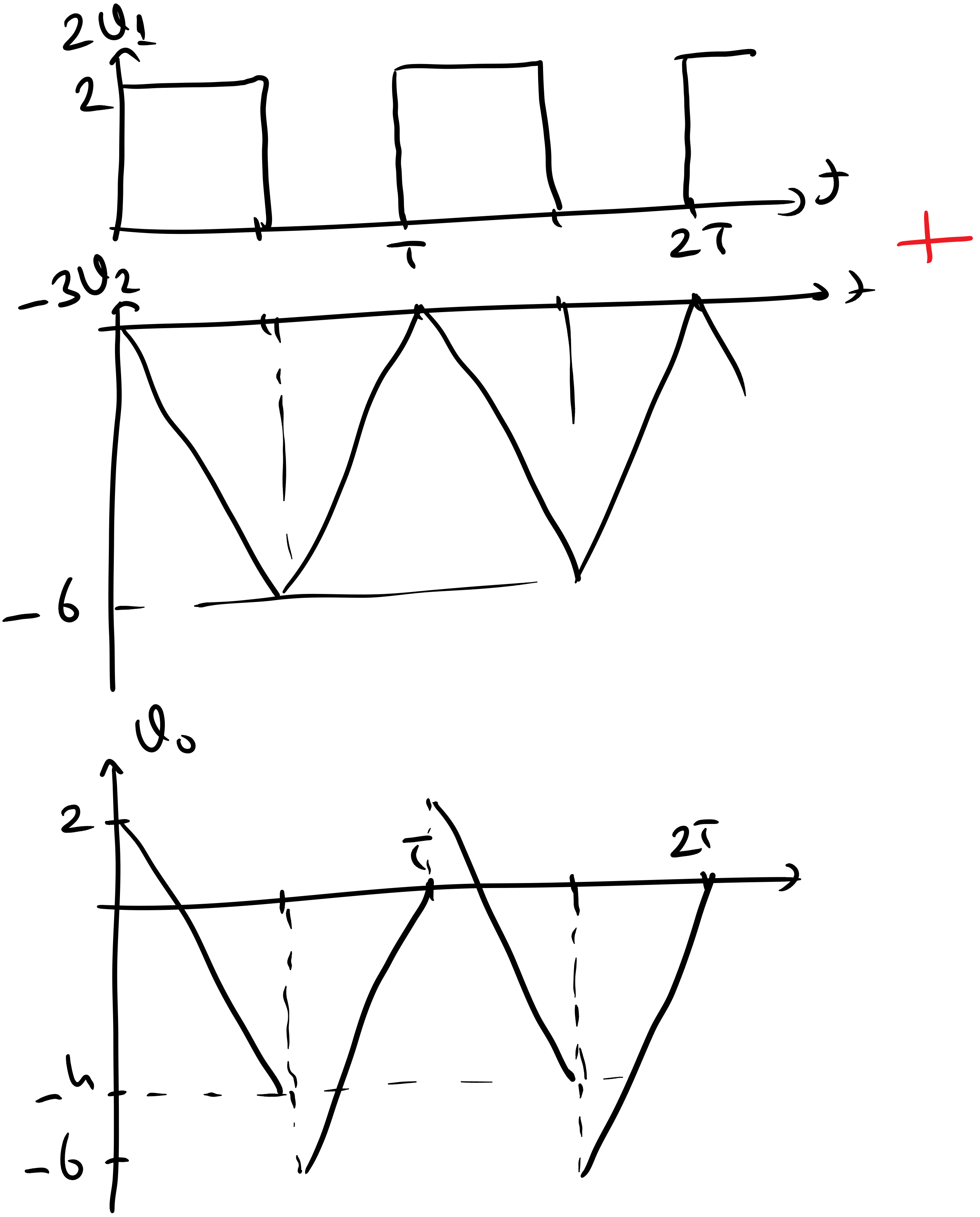
$$-\frac{R_2}{5k} = -3 \Rightarrow \boxed{R_2 = 15k\Omega}$$

$$\frac{5k + R_2}{5k} \cdot \frac{R_1}{5k + R_1} = 2 \Rightarrow \frac{20k}{5k} \cdot \frac{R_1}{5k + R_1} = 2$$

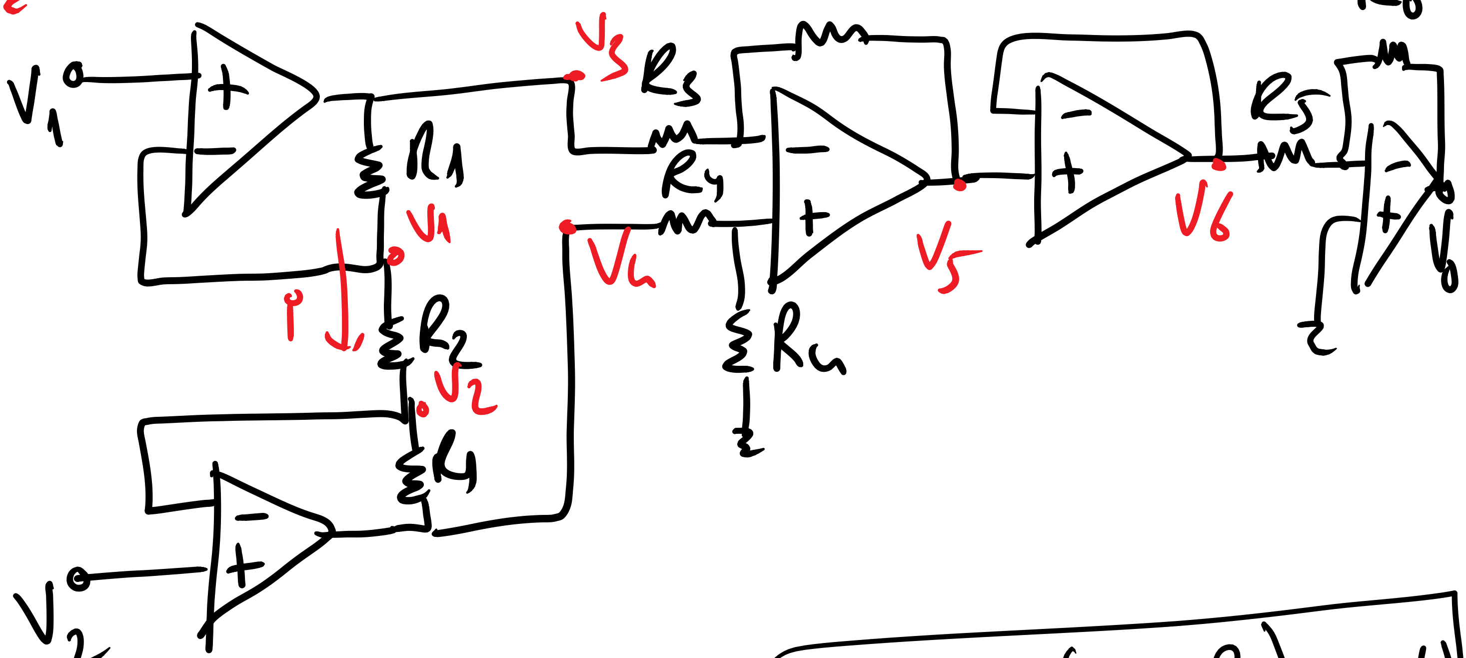
$$4R_1 = 10k + 2R_1$$

$$\Rightarrow \boxed{R_1 = 5k\Omega}$$

b) $v_o = 2v_1 - 3v_2$ 'yi çiz



Örnek-4



$$\frac{V_3 - V_4}{2R_1 + R_2} = \frac{V_1 - V_2}{R_2} \Rightarrow \boxed{V_3 - V_4 = \left(\frac{2R_1 + R_2}{R_2}\right)(V_1 - V_2)}$$

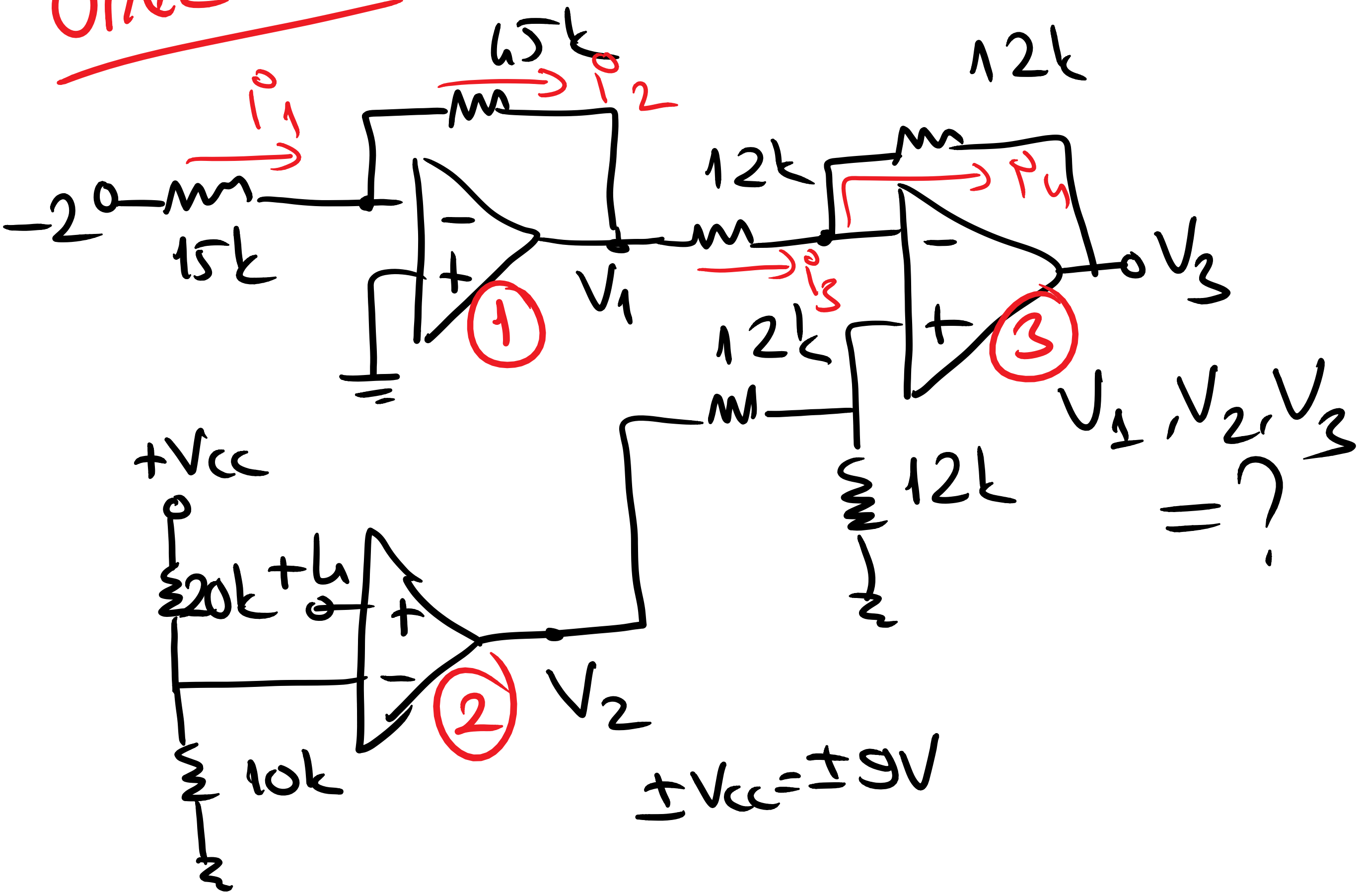
$$V_5 = -\frac{R_4}{R_3}(V_4 - V_3) \Rightarrow \boxed{V_5 = \frac{R_4}{R_3}(V_4 - V_3)}$$

$$\boxed{V_6 = V_5} \qquad V_0 = -\frac{R_6}{R_5} \cdot V_6 = -\frac{R_6}{R_5} \cdot V_5$$

$$V_0 = \frac{R_6}{R_5} \cdot \frac{R_4}{R_3}(V_3 - V_4)$$

$$\boxed{V_0 = \frac{R_6 \cdot R_4 \cdot (2R_1 + R_2)}{R_5 \cdot R_3 \cdot R_2}(V_1 - V_2)}$$

Örnek 5:



1. op-amp'ta:

$$V_1 = -\frac{45k}{15k} \cdot (-2) = 6V \Rightarrow \boxed{V_1 = 6V}$$

2. op-amp'ta: (karşılaştırıcı)

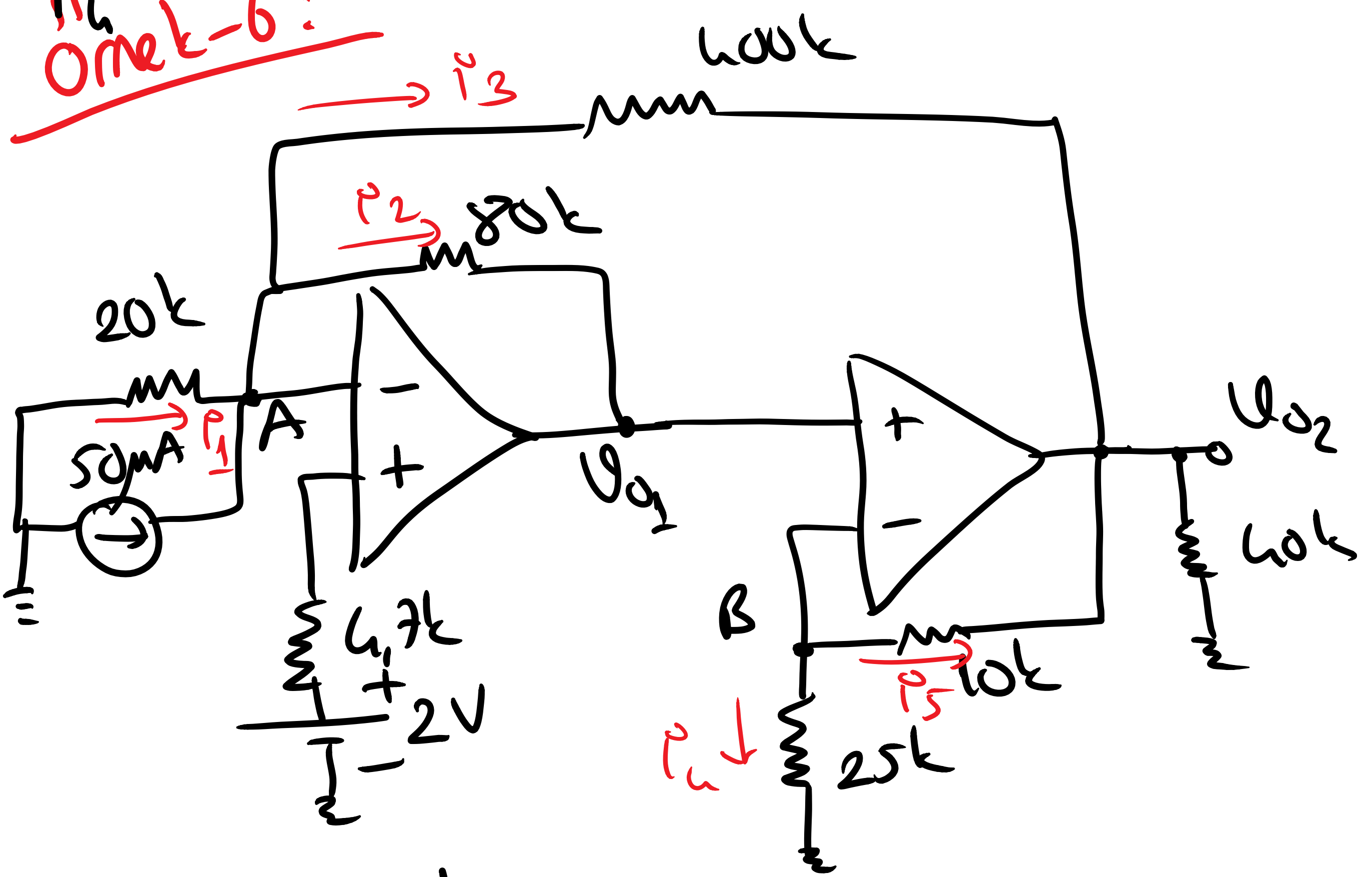
$$V_- = \frac{9}{30k} \cdot 10k = 3V \quad , \quad V_+ = 4V \Rightarrow \boxed{V_2 = 9V}$$

3. op-amp'ta (çıkarma devresi)

$$V_3 = \frac{12k}{12k}(V_2 - V_1) = 9 - 6V = 3V$$

$$\boxed{V_3 = 3V}$$

Örnek-6:



A noktasında:

$$i_1 + 50\mu A = i_2 + i_3 \qquad V_- = V_+ = 2V$$

$$\frac{-2}{20k} + 50\mu = \frac{2 - V_{o1}}{80k} + \frac{2 - V_{o2}}{400k}$$

$$50\mu - 100\mu = \frac{10 - 5V_{o1}}{400k} + \frac{2 - V_{o2}}{400k}$$

$$-50\mu \cdot 400k = 12 - 5V_{o1} - V_{o2}$$

$$\boxed{5V_{o1} + V_{o2} = 32 \text{ ---- } (1)}$$

B noktasında:

$i_4 + i_5 = 0 \qquad V_+ = V_- = V_{o1}$

$$\frac{V_{o1}}{25k} + \frac{V_{o1} - V_{o2}}{10k} = 0$$

$$\frac{2V_{o1}}{50k} + \frac{5V_{o1} - 5V_{o2}}{50k} = 0$$

$$\Rightarrow 7V_{o1} = 5V_{o2} \Rightarrow \boxed{V_{o2} = \frac{7}{5} V_{o1}} \quad ...(2)$$

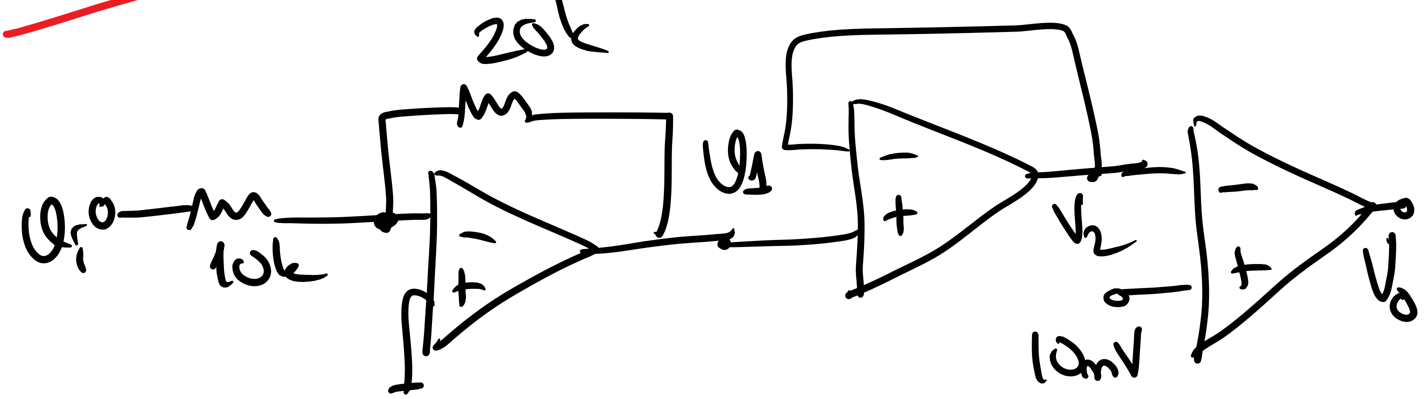
(2) nolu ifadeyi (1) nolu denklemde yerine yazarsak:

$$5V_{o1} + \frac{7}{5} V_{o1} = 32$$

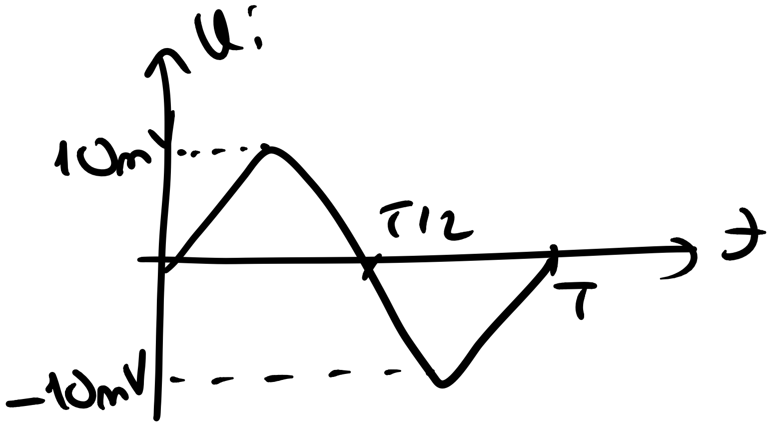
$$\frac{25V_{o1} + 7V_{o1}}{5} = 32 \Rightarrow \frac{\cancel{32}}{5} V_{o1} = \cancel{32}$$

$$\boxed{V_{o1} = 5V} \quad \boxed{V_{o2} = 7V}$$

Örnek-7:



$\pm V_{cc} = \pm 10V$



$V_1, V_2, V_0 = ?$ çiz.

$$V_1 = -\frac{20k}{10k} \cdot U_i \Rightarrow \underline{\underline{V_1 = -2U_i}}$$

$$\underline{\underline{V_2 = V_1}}$$

$$V_0 = \begin{cases} +10 & V_2 < 10mV \text{ ise} \\ -10 & V_2 > 10mV \text{ ise} \end{cases}$$